

Abin Anil

AI/ML Engineer | Python, GenAI, RAG & Agentic Systems

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Professional Summary

AI/ML Engineer with a strong Python foundation, building and deploying GenAI, LLM, and agentic AI applications using LangChain, LangGraph, Hugging Face Transformers, RAG pipelines, and vector search (PostgreSQL/pgvector). Hands-on across the full AI development lifecycle – machine learning theory, deep learning, NLP, model fine-tuning (QLoRA/PEFT), multi-agent orchestration, backend APIs, and cloud deployment – backed by 2+ years of enterprise production-engineering experience (REST APIs, SQL, observability, root-cause analysis) that translates directly into building AI systems that are reliable and production-ready, not just proof-of-concept notebooks.

Technical Skills

Programming: Python, SQL, JavaScript

Machine Learning: Supervised & Unsupervised Learning, Regression, Classification, Ensemble Methods (XGBoost, Random Forest), Feature Engineering, Hyperparameter Tuning, Model Evaluation (Precision/Recall/F1), Scikit-Learn

Deep Learning: Neural Networks, CNNs, RNNs, Transformer Architectures, TensorFlow

GenAI/LLM & Agentic AI: Agentic AI Workflows, Multi-Agent Systems & Orchestration, Workflow Orchestration (LangGraph state graphs), Agent Memory & State Management, Tool Calling / Function Calling (Model Context Protocol – MCP), Retrieval-Augmented Generation (RAG), LLM Application Development & System Integration, Generative AI, Generative AI Agents, Large Language Modeling, LangChain, LangGraph, Hugging Face, Prompt Engineering, Prompt Patterns, Context Engineering, Vector Embeddings, Semantic Search, QLoRA, PEFT, Unsloth, 4-bit Quantization

NLP: NLTK, spaCy, Tokenization, TF-IDF, Sentiment Analysis, Text Classification

Backend/Data: FastAPI, PostgreSQL, pgvector, SQLAlchemy, pycopg, REST APIs, Pandas, NumPy

Cloud/Deployment: Docker, Docker Compose, GCP Cloud Run, Jenkins CI/CD, Git, GitHub

Engineering Fundamentals: API Debugging, Root Cause Analysis, Observability, Incident Management, OOP, Data Structures & Algorithms, SDLC

Projects

Autonomous Multi-Agent Trading Architecture (Nexus Intelligence)

Sep 2026

GitHub: <https://github.com/abinanil000datascience/nexus-ai-trading-engine>

- Architected a fully decoupled, asynchronous multi-agent trading system, replacing a single-threaded Streamlit prototype, by rebuilding the backend on FastAPI and orchestrating four autonomous agents (Scout, Quant, Analyst, Gatekeeper) using LangGraph state-graph workflow orchestration with persistent agent memory across multi-step reasoning.
- Implemented structured tool calling and function calling between agents via the Model Context Protocol (MCP), enabling each agent to invoke shared tools and pass state through a coordinated agentic workflow.
- Enabled sentiment-aware, real-time trading verdicts by combining a deterministic Quant agent – computing live 14-day RSI/MACD on streaming yfinance data via Pandas – with an NLP-driven Analyst agent that uses Gemini 3.5 Flash to classify live financial news as Bullish/Bearish/Neutral.
- Eliminated critical concurrency failures in the agent orchestration layer, resolving asyncio event-loop collisions between FastAPI and LangGraph execution threads, by restructuring API endpoints to run agent workflows on isolated background threads.
- Delivered a production-style monitoring interface for agent decision-making, visualizing 60-day price action, live agent telemetry, and trade verdicts in real time, via a React/Vite/Tailwind CSS v4 dashboard with Recharts against the FastAPI backend.

Tech Stack: Python, FastAPI, LangGraph, MCP, Gemini 3.5 Flash, yfinance, Pandas, asyncio, React, Vite, Tailwind CSS, Recharts.

Autonomous Healthcare ITSM Resolution Engine

Aug 2026

GitHub: https://github.com/abinanil000datascience/Healthcare_ITSM_Agent

- Built an LLM-powered IT ticket resolution system by fine-tuning a local Meta Llama-3 (8B) model with Unsloth and QLoRA (4-bit quantization) for domain-specific ticket classification and generation.
- Implemented a full Retrieval-Augmented Generation (RAG) pipeline, combining fine-tuned Llama-3 generation with hybrid PostgreSQL/pgvector retrieval to ground ticket responses in historical resolution data.
- Eliminated paid cloud LLM API costs by engineering a zero-marginal-cost local inference pipeline for IT ticket triage.
- Accelerated retrieval across a historical ticket corpus by designing a sub-linear hybrid search architecture in PostgreSQL/pgvector, combining keyword matching (tsvector/BM25) with HNSW-indexed semantic vector search.
- Deployed the retrieval system as an asynchronous FastAPI backend using sentence-transformers, fully containerized with Docker.

Tech Stack: Meta Llama-3 (8B), Unsloth, QLoRA, Hugging Face, PostgreSQL, pgvector, HNSW, sentence-transformers, FastAPI, Docker.

Enterprise AI Career PlatformAug 2026

GitHub: <https://github.com/abinanil000datascience/ai-career-platform>

- Built an AI-powered career platform for resume analysis, job matching, and applicant tracking.
- Automated resume-to-role matching by developing a semantic ATS evaluation engine using LangGraph workflow orchestration and vector embeddings, backed by a PostgreSQL and pgvector data layer for retrieval-grounded matching.
- Implemented multi-provider LLM system integration (Gemini API, Groq API) on an asynchronous FastAPI back-end.
- Deployed the containerized service on GCP Cloud Run.

Tech Stack: Python, FastAPI, LangGraph, Gemini API, Groq API, PostgreSQL, pgvector, Docker, GCP Cloud Run.

AI-Powered Supply Chain Risk PredictorMay 2026

GitHub: <https://github.com/abinanil000datascience/supply-chain-risk-predictor>

- Forecasted freight delays on a 97% on-time, class-imbalanced dataset by building a predictive pipeline (Scikit-Learn, XGBoost) with weighted models to correct for imbalance.
- Delivered delay-risk predictions to end users in real time through a Django REST API backend with a React front end, giving visibility into at-risk shipments.

Tech Stack: Python, Scikit-Learn, XGBoost, Django, React.

Professional Experience

Technical Support Engineer, FeathersoftAug 2024 – Present

Distributed DNA diagnostics platform (DDC)

- **API & Database Operations:** Managed production reliability for a distributed DNA diagnostics platform (DDC), ensuring data integrity across complex probability (PI) calculations and automated report generation workflows, sustaining 99.8% system uptime.
- Reduced production error rates by 22% by debugging API-layer failures and data-ingestion bottlenecks through network traffic inspection, cURL requests, and payload validation using Postman and Swagger.
- Spearheaded system observability and anomaly detection by analyzing application logs in Kibana and Grafana to isolate root causes of complex workflow disruptions, accelerating incident resolution time by 15%.
- Reduced Mean Time to Resolution (MTTR) by 20% by executing direct SQL (SELECT/UPDATE) database interventions to resolve live data discrepancies and maintain the structural integrity of diagnostic reporting.
- Engineered and validated temporary diagnostic workflows within staging and test environments, acting as a technical bridge between end-users and the core development team to unblock production.
- Led cross-functional incident triage with US-based enterprise clients and engineering teams, translating complex diagnostic logs into actionable technical fixes, sustaining a 99% CSAT rating across 120+ escalated incidents.

Education

Bachelor of Technology (B.Tech) in Computer Science and EngineeringDec 2020 – May 2024

University College of Engineering, ThodupuzhaCGPA: 7.33/10.0

Certifications & Training

- **Fundamentals of AI Agents Using RAG and LangChain** – IBMSep 2026
Credential ID: S0HTSWVNLSTW | [Verify Credential](#)
- **Data Science With AI Professional Certification** – MetaiLearnNov 2025 – Jun 2026
- **Python Programming & Machine Learning Intensive Program** – Reverttech IT Solutions (1 month)Dec 2024
- **Advanced Machine Learning Workshop** – Techmaghi Training InstituteSep 2022

Achievements

- **Employee of the Month (January 2026)** – Recognized for technical excellence, critical root-cause analysis, and outstanding client communication.